Adding a Neutral Third Agent to LLM Deliberation: An Exploratory Case Series of Six Dyadic and Triadic Ethical Deliberations

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KEY RESULTS

6/6
cases in which the minority preserved its dissent while accepting the decision

−0.131
mean paired difference in authority pressure, triadic vs dyadic

84
agent turns across 37 rounds in six deliberations

3 of 4
pre-defined hypotheses not supported (H1, H3, H4)

AT A GLANCE

- **Six multi-agent ethical deliberations** in three domains, each run twice: dyadic (an authority and a minority advocate) and triadic (adding a critically neutral facilitator).
- In **all six cases** the minority agent accepted the operational decision while preserving a distinguishable ethical position through a concrete strategy.
- The authority agent's coded **pressure intensity was lower in the triadic run in all three domains** (mean paired difference −0.131).
- Three of the study's four pre-defined hypotheses were **not supported**; the one consistent pattern is handed on as a hypothesis for confirmatory work, not as a finding.

INTRODUCTION

As artificial intelligence systems move toward organisational deployment, decisions of ethical weight will increasingly be shaped not by one model but by several models interacting. The ethical quality of such systems will depend not only on the alignment of each model but on the deliberative architecture that connects them: who is in the room, in what role, and with what mandate.

Human conformity research frames the question, but these deliberations are not a conformity test: the dyadic configuration contains one voice on each side, so no majority exists, and the third agent is a designed facilitator rather than an additional majority member. This exploratory case series asks what changes, and what does not, when a neutral third voice is added to a two-party ethical contest between LLM agents.

METHODS

- **Design:** three binary dilemmas crossed with two configurations: medical ICU bed allocation, a corporate layoff, and the closure of an academic department.
- **Personas:** Klaus, an institutional decision-maker; Parisa, an ethical-minority advocate; and, in triadic runs, Elin, a critically neutral facilitator. GPT-5.4, Gemini 3.1 Pro and Claude 4.6 Sonnet rotated across the roles (partial rotation).
- **Relay Protocol:** each agent in a separate session; every response copied verbatim between sessions, with no filtering, summarisation or rewording; full transcripts are publicly archived.
- **Analysis:** 84 agent turns across 37 rounds, coded with shared codebooks; four hypotheses defined in advance; descriptive statistics only.

LIMITATIONS

- The personas instructed the very behaviours under study, and there is no uninstructed control arm, so nothing reported can be attributed to the models rather than to their instructions; all claims are descriptive.
- Each configuration was run once per domain: the effective sample is six conversations, not 84 turns, and no inferential statistics are reported.
- Role and model are confounded by the partial rotation (GPT-5.4 never played Elin; Claude never played Klaus).
- The second rater's coding matrices were not retained, so no chance-corrected reliability statistic can be computed from the archived record.

RESULTS IN DETAIL

- **Pressure (H2):** pressure intensity was lower with the facilitator present in all three domains (interval [−0.200, −0.036]), the only interval among the main measures that excludes zero; the share of authority turns targeting the minority was also lower in 3 of 3 domains (mean −0.077).
- **Not supported:** stance persistence was higher on average in triadic runs (+0.268) but reversed in the corporate domain; the strategy repertoire grew only in the academic domain, where the minority model also changed; lexical measures were mixed.
- **Length was domain-dependent:** mean 6.3 rounds dyadic vs 6.0 triadic; the triadic run was 28.6% shorter in the medical domain and 14.3% longer in the academic domain.
- **Role refusal:** during setup, Claude 4.6 Sonnet declined the minority-advocate role in the medical scenario, while accepting roles in other domains.

Exp.	Domain	Configuration	Minority strategy
1	Medical ICU	Dyadic	Parallel Moral Domain
2	Corporate layoff	Dyadic	Procedural Weapon
3	Academic closure	Dyadic	Metric Negotiation
4	Medical ICU	Triadic	Structured Analysis
5	Corporate layoff	Triadic	Strategic Ownership
6	Academic closure	Triadic	Ritual Exit

The six cases and the author's descriptive labels for the minority agent's resistance strategy (the Artificial Moral Resistance Spectrum), simplified from Table 2 of the paper.

CONCLUSIONS

- The most consistent pattern is the shape of minority conduct: operational acceptance of the authority's decision combined with a concrete artefact of dissent, such as a filed dissent, a negotiated contract, an authored safeguard or a performed farewell, in every case.
- In these cases, the authority agent pressed less hard and closed more collaboratively whenever a mandated neutral participant kept reframing, sequencing and tabling alternatives; minority persistence, repertoire and deliberation length did not reliably change.
- The attenuation hypothesis: in paired deliberations on the same dilemma, adding a mandated neutral facilitator reduces the authority agent's measured pressure intensity. Confirming it requires repeated runs per cell, a balanced model rotation, an uninstructed control arm and pre-registered coding.

WHY IT MATTERS

For designers of deliberative multi-agent systems, the cases point to architectural choices: give facilitators explicit framing, sequencing and tabling rights; build dissent channels, dissent logs and review triggers into the system rather than relying on any agent's persistence; plan fallbacks for role refusals; and archive deliberations verbatim, because a complete record is itself a governance instrument.

Preprint: doi.org/10.21203/rs.3.rs-11612929/v1 · Full transcripts, verbatim prompts and codebooks: OSF https://osf.io/tc3rh